AI UX & Data Visualisation Design Principles (CA6002)

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Chapter 5.1 - HCAI UX Design - Usability

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Designing Usability

Why is Usability Important?

- **Usability** is a measure of **how well** a specific user in a specific context **can use** a product/design to achieve a defined goal. High usability implies that this goal can be achieved **intuitively**, **effectively**, **efficiently** and **satisfactorily**.
- Usability in AI applications is crucial for **driving adoption** and **building stickiness** in the product (i.e. customers return to the product/service repeatedly).
- E.g. Microsoft's Copilot can now integrate with user's **familiar** Word application to provide **seamless** and **contextual** assistance to enhance productivity **without disrupting** user's workflow.

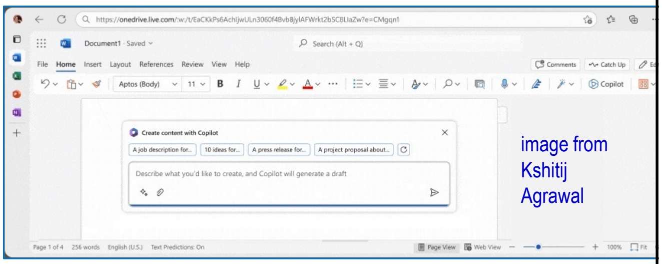
image from Kshitij Agrawal

[1] Interaction Design.org, What is usability? - https://www.interaction-design.org/literature/topics/usability
[2] Kshitij Agrawal, The Ultimate collection of Generative AI-UX Interactions, Medium - https://uxplanet.org/the-ultimate-collection-of-generative-ai-ux-interactions-6367adbf2ad1

Designing Usability

Usability Design

- Usability is not equivalent to user experience (UX) but a **2nd level component of UX design** as it comes after the **utility** (i.e. the ability to solve user's problems) but before desirability. Thus, ensuring the AI system works as intended is still the first priority.
- A product's usability is **by design** and should contain the following elements:
  - **Effectiveness** - supports users in completing actions **accurately**.
  - **Efficiency** - Users can perform tasks **quickly** through the easiest process.
  - **Engagement** - Users find it **pleasant** to use and appropriate for its industry/topic.
  - **Error tolerance** - It **accommodates** a wide range of user actions and **feedback errors intelligibly** in genuine erroneous situations.
  - **Ease of learning** - New users achieve goals **easily & improves** quickly with use.

[1] Interaction Design.org, What is usability? - https://www.interaction-design.org/literature/topics/usability

Designing AI Interactions

Design Guidelines for Intuitive AI Interactions

- Many elements of usability design such as effectiveness, efficiency and easy of learning are addressed by **interaction designs** that are **intuitive** and **easy to use**.
- Interaction design involves the design of both what the AI systems **shows the user** (e.g. high awareness of the current state of the system) and how the **user can respond** or specify the desired operations (e.g. which button should I press, how should I input the info?).
- Some of the relevant issues related to designing intuitive AI interactions include.
  - **Consistency** in the user interface and its deployment across delivery platforms.
  - Design considerations for **new users**.
  - **Awareness of AI** enhancements in the interaction.
  - **Informative system feedback** after user action.

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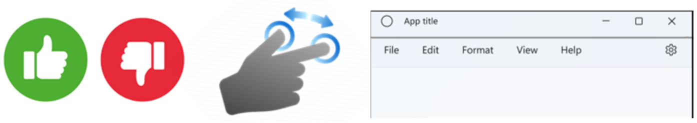
[1] Interaction Design.org, What is usability? - https://www.interaction-design.org/literature/topics/usability

Designing AI Interactions

Design Guidelines - Consistency

- **Consistency** in design is the key to creating **intuitive** AI interfaces. Established design patterns, conventions and familiar visual elements allow users to navigate and interact easily using **existing mental models** built from previous experiences.
- Use **familiar** icons (e.g. thumbs down/up for –ve/+ve feedback), interactions patterns (e.g. pinch in/out to zoom in/out of map) and layout (e.g. menu bar on top).



- User interface design (e.g. colour scheme, menu item order, content order, etc) should be consistent **across** different **delivery platform** (e.g. web and mobile app).

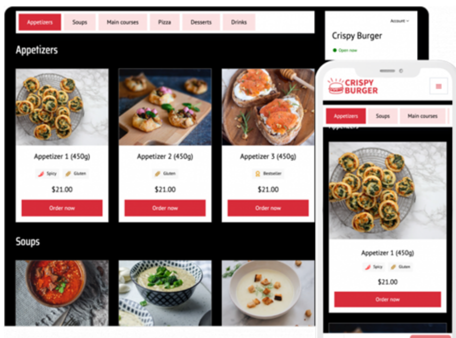


image from upmenu.com

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[3] Rupa Chaturvedi, Design human-centered AI interfaces, Reforge.com - https://www.reforge.com/guides/design-human-centered-ai-interfaces
[4] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

Designing AI Interactions

Design Guidelines - Tooltips for Newbies

- User interface (UI) design is considered good when **new users** can **find their way** through the application without problems and **quickly achieve their goals**.
- UI designs like **tooltips** provide **contextual help** (e.g. pop-up text describes what icon does when cursor hovers over it) can make it easier for unfamiliar users to navigate through the app features and its operation.
- Informative tooltips are useful initially but becomes distracting once familiar. Allow users to disable such tooltips (e.g. by clicking the **"Don't show.."** button).

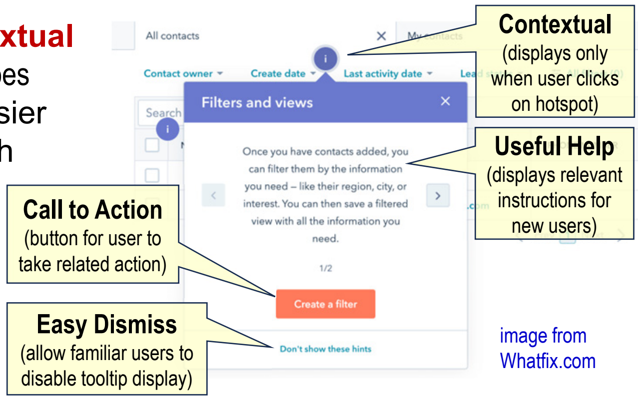

image from Whatfix.com

[5] A. Siarkiewicz and D. Sobolewski, Applying the 8 Golden Rules of User-Interface Design, UXmatters.com
[6] Levi Olmstead, How to Create & Use UI Tooltips (+Examples), WhatFix.com, 24 Dec 2024 – https://whatfix.com/blog/tooltips/

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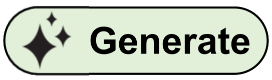
Designing AI Interactions

Design Guidelines - Awareness of AI

- Subtle **visual cues** can impact user experience of an AI-powered app as they allow them to seamlessly adapt and embrace its **AI features** (e.g. in chat interfaces, AI visual cues help users identify and differentiate AI-generated responses from that of a human agent).
- The Sparkles icon (✨) is often used to identify AI. This makes it a useful design pattern that will give users a recognisable AI visual cue that can create awareness of AI enhancements in the user interaction.



- Established AI service providers (e.g. Google Gemini, Microsoft Copilot, etc) ensure that their well- recognised AI visual cues (i.e. their AI icons) are used consistently in apps where their AI features have been integrated.

Copilot Gemini Meta AI ChatGPT

[7] Tetiana Sydorenko, AI visual language: How to help users spot AI-powered features, UX Collective, 8 May 2024, https://uxdesign.cc/ai-visual-language-how-to-help-users-spot-ai-powered-features-3272a1f77d44

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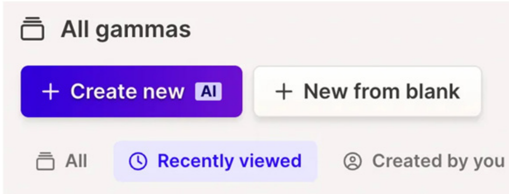
Designing AI Interactions

Design Guidelines - Awareness of AI (more)

- Text badges with **short labels** can also be used to provide contexts that identifies AI features in the application (e.g. the text "AI" appears in the button activating AI functions).
- In terms of **colour palette**, popular choices identifying AI products/features appear to be **purple** and then **green** (But there are no established standards on such colour conventions).



- While incorporating AI visual cues into the user interface design can improve user experience, one must be mindful of the **varying interpretations** of visual elements like symbols and colour across **cultures** and **societies**. User studies across the different demographics may be needed to better gauge such differences before AI product deployment.

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[7] Tetiana Sydorenko, AI visual language: How to help users spot AI-powered features, UX Collective, 8 May 2024, https://uxdesign.cc/ai-visual-language-how-to-help-users-spot-ai-powered-features-3272a1f77d44

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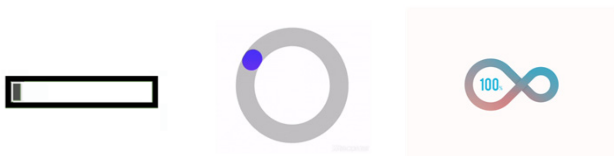
Designing AI Interactions

Design Guidelines - Informative System Feedback

- A well-design AI system should be **responsive** to users' action and keep them **informed** about the **results of their action** (i.e. users want to understand if their action has been registered and being processed by the AI algorithm).
- The system response should be **immediate** and **understandable** to the users. If the results will take some time to return, users should be kept engaged with the **assurance** (e.g. via some animated display) and shown a **duration estimate** (e.g. via a progress bar) of when the results is likely to appear (i.e. if that duration can be estimated).
- Such feedback will help **maintain trust** in the AI system as **no visible feedback** after user's action will give the impression the system is **broken** and not working.



Examples of animated progress visuals

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[5] Anna Siarkiewicz and Dawid Sobolewski, Applying the 8 Golden Rules of User-Interface Design, UXmatters.com, Oct 2022 - https://www.uxmatters.com/mt/archives/2022/10/applying-the-8-golden-rules-of-user-interface-design.php

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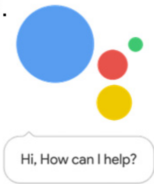
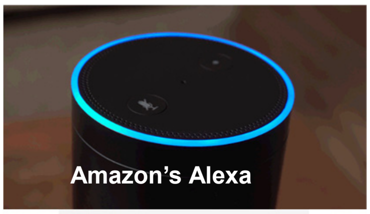
Designing AI Interactions

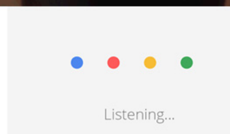
Design Guidelines - Informative System Feedback (more)

- Information feedback is very important for **voice user interface** (e.g. voice assistant (VA) like Alexa or Google Assistant). Users need to be aware when the VA is **actively listening** and they should speak (e.g. Google Assistant indicates this using animated dots and Alexa uses a bluish swirling light at the top of its rim).
- When the device is trigger to listen, a **brief & transitory** cue (e.g. "*Hi! how can I help*?") should prompt **immediately** to affirm start of conversation.





- Confirmation of task done will give users **assurance** the AI system is **competent** and the **command understood** (e.g. a command like "*Switch off my dining lights*" must come with a VA response like "*Dining lights turned off*", when task is completed).



Google Voice Assistant

[8] P. Chawla, How-to Guide for a Flawless Voice User Interface Design, 20 Dec 2024 - https://appinventiv.com/blog/voice-user-interface-design/

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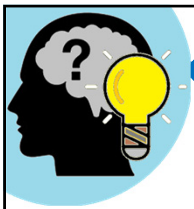
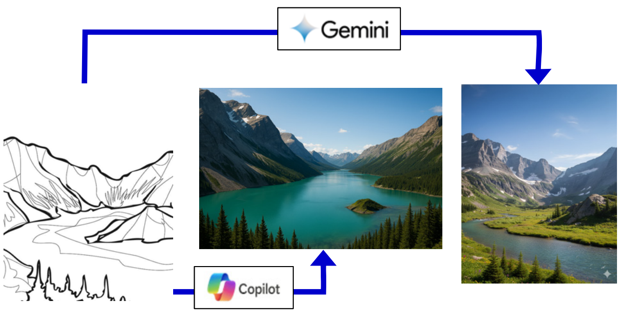


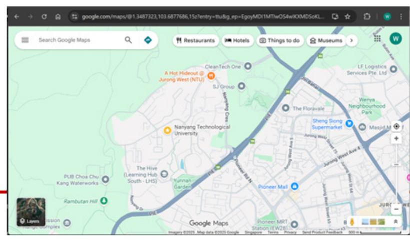
Think and Apply

Designing AI Interactions

- How do Google Gemini and Microsoft Copilot design awareness of AI and informative system feedback when we get them to convert line drawings photo-realistic colour images?



- Are there different tooltip designs in Google Maps? How are they different and what is the rationale for their different designs?



Google Maps

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Designing Error Handling

Design Guidelines for Handling Errors

- AI systems are **probabilistic** in nature. As such, **errors** and **inaccuracies** are bound to occur and AI systems should be **designed** with the understanding that these errors are an integral part of the **user experience**.
- **Errors** and **failures** must be distinguished. What users consider an error is related to their expectation of the AI system (e.g. a movie streaming service that predicts an uninteresting recommendation can be viewed as an error, but an autonomous vehicle that fails to stop at a traffic light is a system failure).
- The line between the two is sometimes **fuzzy** (e.g. when a music app does not respond to a user's action, that may be viewed as a failure, and an irrelevant song prediction is an error). Our design should **tolerate errors** but **not failures** as the latter make the system **unusable** while errors may just result in mere **inconvenience**.

[9] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022
[4] Ben Shneiderman, Human-Centered AI, Oxford University Press, 2022

Designing Error Handling

Types of Errors

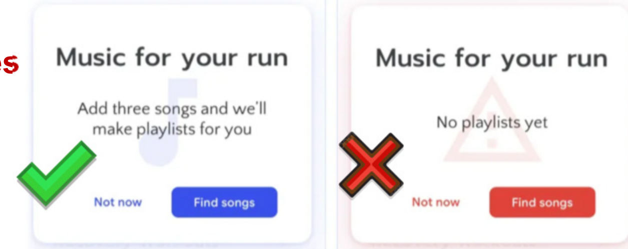
- There are two basic types of errors that may be encountered in an AI system or during interaction with the system, namely **AI system error** and **user error**.
- AI **system errors** could occur due to the **inherent limitations** of the AI system resulting from inadequate training (i.e. insufficient data, poor quality data or biasness in the data), incorrect AI model/algorithm selection, improper emphasis applied to the different input features, etc.
- **User errors** happen when users **make mistakes** interacting with the AI (e.g. incorrect or ambiguous inputs, pressing wrong button, etc).
- However, the worst kind of error that **erodes user trust** is when users make mistakes (and they are aware of it), but the AI system gives no feedback or meaningful response. This is why good **error handling** is an important HCAI UX design.

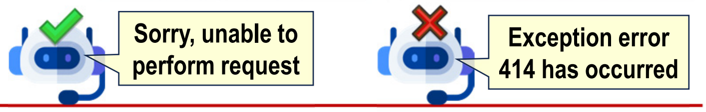
[9] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

Designing Error Handling

AI System Error - Effective Communication

- AI systems should effectively **communicate** error states and provide **explanations** that can help established a **better mental model** of how the AI system works (e.g. instead of saying an uninitialised music recommendation system has "*no playlist yet*", say "*Add 3 songs and we will create a playlist for your*").
- Such error states provide **opportunities** to **guide users** on how to improve the AI's performance and develop better a understand of how AI systems learn.
- Explanation of error states should be in **understandable** form (e.g. say "*Sorry, I am unable to help*" instead of "*Bad input: 414 exception error.*").





[10] Sarah Tan, How to Build Human-Centered AI Products that Build Trust with Your Users, Medium article, 21 Jul 2023 – https://medium.com/@sarahtan.twh/cracking-the-code-of-trust-how-transparency-revolutionizes-human-centered-ai-experiences-de8748b40067

Designing Error Handling

AI System Error - Opportunity for User Feedback

- Misclassification **errors** by the AI system are opportunistic **contextual** moments to solicit **user feedback** to improve the performance of the system (e.g. a dog detection app allows users to provide a simple 👍 or 👎 feedback if the prediction has misclassified, e.g. the well-known a blueberry muffin for a chihuahua)
- AI predictions can be **irrelevant** and **insensitive** (e.g. A ride-hailing service recommends a vacation stay when the user has booked a taxi ride to a hospital), leading to a mismatch of user expectations (which erodes trust).
- **Explicit user feedback** can help the AI system reduce such results in future.

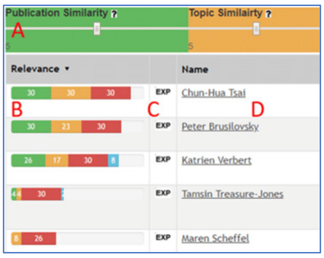
image from Akshay Kore

[9] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

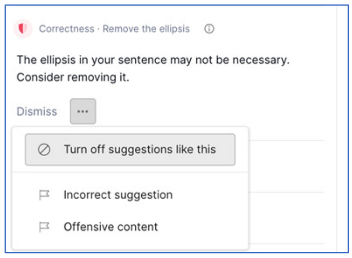
Designing Error Handling

AI System Error - User-perceived Errors

- Sometimes AI results feel like a system error (i.e. user-perceived error) to the user but it is giving optimal results based on the data and settings used by the AI system.
- Such **context errors** occur due to the **incorrect assumptions** by the AI about the **user preferences** (e.g. A viewer watches a horror movie because of an actor. Horror movies starts being recommended due to the AI's high weightage settings on "movie genre").



Weightage control (Relevance Tuner+)



Allow explicit feedback (Grammarly)

- Methods to ameliorate such errors include giving **more control** over weightage used (e.g. in Relevance Tuner+) and allowing explicit **user feedback** (e.g. in Grammarly).

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[9] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022
[11] C.Tsai, P. Brusilovsky, The effects of controllability and explainability in a social recommender system, UMUAI 31(4):1-37, 2021

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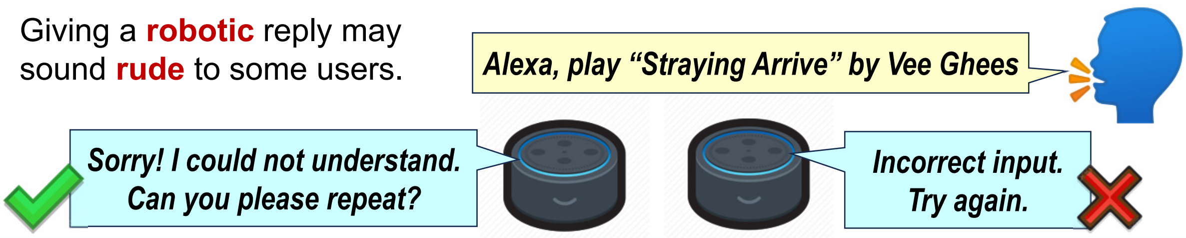
Designing Error Handling

User Error - Don't Blame the User

- Users will make mistakes during their interaction with the AI system. Avoid error communications that suggest or imply that the **user is incompetent** or is doing something wrong (e.g. don't say "*illegal command*" or "*your input is incorrect*").
- Phrase messages **politely** (e.g. say "*we are unable to process this right now*" or "*can you check the information entered and try again*"). Instead of accusing user of an error, indicate error has occurred and provide **helpful advice** on what caused it and how it can be fixed.
- Giving a **robotic** reply may sound **rude** to some users.



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[9] Akshay Kore, Designing human-centric ai experiences: applied UX design for artificial intelligence, Apress, 2022

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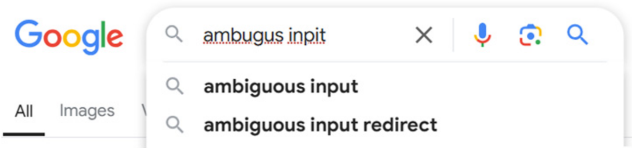
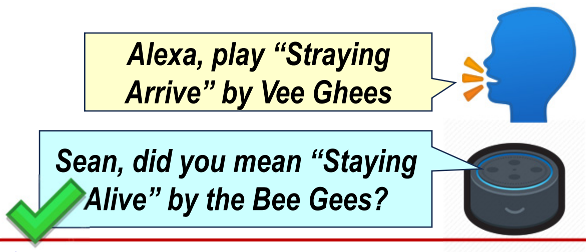
Designing Error Handling

User Error - Handling Ambiguous Inputs

- AI interfaces should gracefully handle ambiguous inputs by **offering viable suggestions** that represent **alternative interpretation** of user's uncertain input (e.g. by saying "*did you mean …*").
- Offer user assistance to avoid erroneous inputs by providing a range of possible correct input options (e.g. in the form of an **auto-corrected** list of **N-best** possibilities). This anticipation helps **prevent errors** in the first place.



Google search auto-corrects as users input their text & displays a list of viable options



[3] Rupa Chaturvedi, Design human-centered AI interfaces, Reforge.com - https://www.reforge.com/guides/design-human-centered-ai-interfaces

Designing Error Handling

User Error - Preventing Errors

- Another good design technique to prevent errors is to include **helpful constraints**, especially if there are clear rules for the data input (e.g. date selection widgets for specifying a date like the DOB or a time period like flight booking dates where depart > return).
- **Error prevention tooltips** can be used to alert users to potential errors or pitfalls before they occur (e.g. guide on correct email entry by detecting typical email formats).

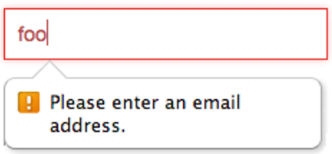


image from Stack Overflow

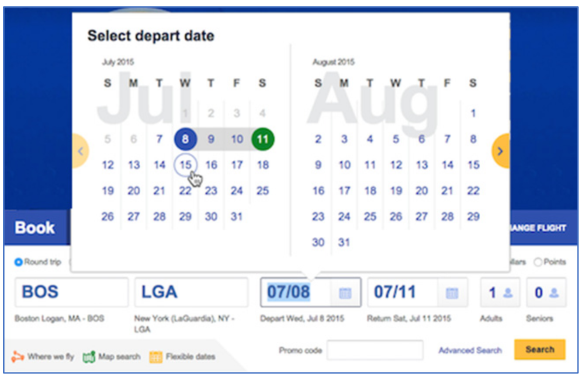


image of Southwest Airlines date selection modified from Laubheimer

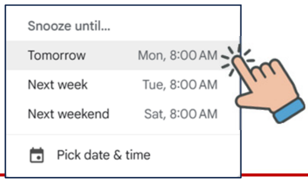
[12] Page Laubheimer, Preventing User Errors: Avoiding Unconscious Slips, 23 Aug 2015 - https://www.nngroup.com/articles/slips/

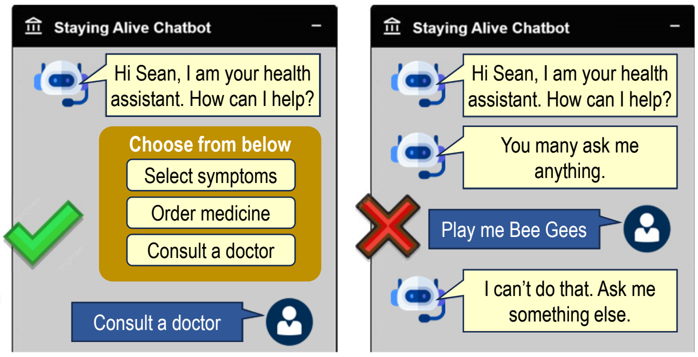
Designing Error Handling

User Error - Preventing Errors (more)

- Unnecessary errors due to wrong expectations can be prevented by highlighting the **capabilities** and **scope** of the AI services **upfront** (e.g. a Health chatbot with selectable categories to give users a mental model of what the AI can do instead of saying "*Ask me anything!*").
- Where possible, give **good default** selection options to users (e.g. reminder app defaults like *1-hour*, *tomorrow*, *next week*).

Good defaults **reduce mistakes** & **teach** users what are **reasonable values** for the question.





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HCAI UX Design - Usability

Summary

- Usability design is about making the AI product **easy** and **efficient** to use, both for **new** and **experience users**.
- Effective AI interactions is facilitated by the principle of **consistency**, users being **aware** of the **AI-elements** in the interaction and users receiving **timely information feedback** after taking action.
- Handling unavoidable **AI system errors** is an integral part of HCAI UX design and should be viewed as an **opportunity** to **explain** to users how they can make the AI work better and get their **feedback** to help improve the AI's performance.
- **Preventing** users from making **mistakes** during interaction with the AI system should be viewed as a responsible part **usability design** and minimising such errors will help improve the overall user experience.

References for Chapter 5.2 - HCAI UX Design - Usability

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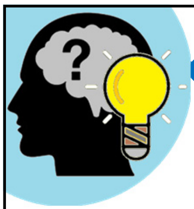
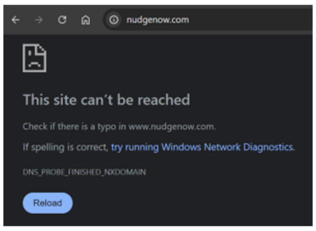
Note: All online articles were accessed between Dec 2024 to Jan 2025



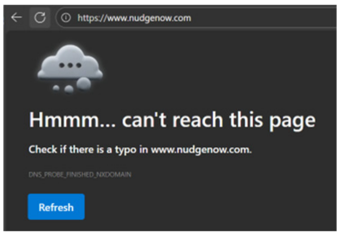
Think and Apply

Designing Error Handling

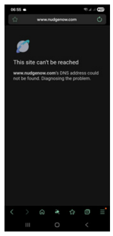
- How do different web browsers handle non-existent websites? Can we learn from the way they have design error handling?
- How do different web sites handle missing pages (i.e. the 404 Error)?
- Can we learn from the way they tackled some error handling issues to help us improve our HCAI UX usability design?



Google Chrome



Microsoft Edge



Samsung Internet

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Chapter 5.1 - HCAI UX Design - Usability

The End

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